Journal of English for Academic Purposes 74 (2025) 101485

Contents lists available at ScienceDirect

Journal of English for Academic Purposes

journal homepage: www.elsevier.com/locate/jeap

Local grammar, discourse acts, and disciplinarity: A case study of definition

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ARTICLE INFO

Handling Editor: Dr Hilary Nesi

Keywords:
Local grammar
Discourse acts
Disciplinary variation
Definition

ABSTRACT

This study employed the local grammar approach to explore disciplinary variations in the use of the discourse act 'definition' in research articles in Linguistics, Education, Physics, and Civil Engineering. The study aimed to unveil the intricate connection between language use and disciplinarity, and to offer an alternative approach to examine disciplinary discourses. Four corpora, i.e., LinDEAP, EduDEAP, PhysDEAP, and CivDEAP from the DEAP (Database of English for Academic Purposes) Corpus, were used for the study. Instances of definition were retrieved by searching for a set of lexico-syntactic markers, and the subsequent local grammar analyses of these instances identified 37 relatively frequent patterns associated with definition. We then deliberated on the similarities and differences in the distribution of local grammar patterns of definition between the soft and hard sciences as well as between the pure and applied disciplines. It was argued that the variations in the use of definition not only reflect but also construct the disciplinary epistemologies. The implications for EAP/ESP writing research and pedagogy were also discussed.

1. Introduction

Since the publication of John Swales' *Genre Analysis* in 1990, the variation of language use across different disciplinary contexts has been brought to the forefront of EAP/ESP writing research (Jiang & Hyland, 2020; Liu et al., 2023). The notion of 'discipline specificity' (Hyland, 2002) sees disciplines as discourse communities characterised by unique methods of inquiry, epistemological values, and modes of argumentation. These attributes, which govern the ways disciplinary knowledge is constructed and disseminated, constitute what we refer to as disciplinary culture, or disciplinarity. Differences across disciplinary cultures can be teased out by examining the linguistic forms and rhetorical characteristics of disciplinary discourses. This is because, as Becher (1987) insightfully observes, if different knowledge fields have their distinctive characteristics, "one would expect differences in fields of knowledge to be reflected in differences in linguistic form; and by the same token, differences in linguistic form to signify differences in fields of knowledge" (p. 261).

Numerous efforts have thus been made to explore the disciplinary variations of language use from diverse perspectives. For instance, studies have examined the variations in rhetorical move structures of academic genres across disciplines (Hu & Liu, 2018; Hyland, 2000). Then, with the growing impact of corpus linguistics, a multitude of studies have documented how disciplinary writers

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https://doi.org/10.1016/j.jeap.2025.101485
Received 3 September 2024; Received in revised form 5 February 2025; Accepted 7 February 2025
Available online 8 February 2025
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differ in their deployment of a range of lexico-grammatical or phraseological features, such as metadiscourse (Hyland, 2005; Hyland and Jiang, 2021), lexical bundles (Cortes, 2004; Hyland and Jiang, 2018a), and syntactic complexity (Lu et al., 2021). Additionally, efforts have also been made to probe into the deployment of appraisal resources in disciplinary discourses (e.g., Deng et al., 2024; Zolfaghari, 2023). By employing different theoretical and/or methodological frameworks, these studies have not only enriched our understanding of the linguistic and rhetorical characteristics of disciplinary discourses but also unveiled the complex interplay between language use and disciplinary cultures.

Furthermore, cross-disciplinary studies are of particular relevance to EAP writing pedagogy. Recoginising the specificities of disciplinary discourses could guide EAP practitioners to develop more efficacious and discipline-specific pedagogy, which could in turn empower novice/student writers to master the norms, conventions, and rhetorical strategies specific to their disciplines, thereby facilitating them to acquire discursive competence as disciplinary insiders.

Notwithstanding the value of previous investigations into disciplinarity, it has to be noted that the extant literature has largely focused on the macro-structural or lexico-grammatical aspects of disciplinary discourses, leaving the discursive or pragmatic features such as the performance of discourse acts or functions thinly researched. As Liu et al.'s (2023) bibliometric analysis indicates, EAP research in the past two decades has witnessed a gradual move towards 'a more socially situated and discursively oriented perspective' on disciplinary discourses (p. 5), with a noticeable shift of attention from the linguistic forms to the pragma-discursive functions. Aligning with this trend, a few studies have attempted to use the local grammar approach, an emerging function-oriented approach in corpus linguistics, to explore academic writers' pragmatic performance of a series of discourse acts that are central to academic writing, such as exemplification (Su & Zhang, 2020), definition (Zhang & Su, 2022), and graphic data commentary (Zhang et al., 2024). However, to date, no efforts have been made to explore how the use of such discourse acts may vary according to disciplines.

To bridge the gaps, the current study presents an exploratory study that adopts the local grammar approach to explore how the use of definition varies across four disciplines representing two dimensions of disciplinary distinctions (i.e., hard *vs* soft and pure *vs* applied, see also Section 2.1), aiming to further unveil the association between language use and disciplinarity. The discourse act of definition, according to Dalton-Puffer (2013), has the specific communicative intent of telling others 'about the extension of an object of specialist knowledge' (p. 234). It is particularly prevalent in scientific discourses and across various academic genres (Swales & Feak, 2012). In this study, we follow Zhang and Su (2023) and consider definition as a discourse act to clarify abstract notions or concepts and/or to elaborate technical terms. We focus on definition, on the one hand, because it constitutes "a core function in academic writing" (Triki, 2019, p. 115) which is essential to the construction and dissemination of disciplinary knowledge (Clarence & McKenna, 2017), and on the other, because it is an important analytical skill, which could deepen students' understanding of the central concepts of their subjects, thereby contributing to the mastery of subject matter (Dalton-Puffer, 2007) (see Triki (2019) and Zhang and Su (2022) for more comprehensive reviews). Specifically, this study seeks answers to the following research questions.

1) What are the local grammar patterns of definition in disciplinary discourses?
2) How are the local grammar patterns of definition distributed across soft and hard disciplines and what might the distribution suggest about disciplinary cultures and epistemologies?
3) How are the local grammar patterns of definition distributed across pure and applied disciplines and what might the distribution suggest about disciplinary cultures and epistemologies?

2. Literature review

2.1. Academic tribes, disciplinarity and disciplinary discourse studies

Becher and Trowler (2001) described disciplines as academic *tribes* with differing views, norms, and research practices. In exploring the distinctions between disciplines, Becher (1987, 1994) suggested a two-way categorisation of disciplines into hard *vs* soft and pure *vs* applied ones. According to Becher (2012), hard disciplines are grounded in solid theoretical structures, addressing universal or near universal laws or principles. Within hard disciplines, knowledge accumulates in a cumulative manner, leading to findings that can be widely generalised. Soft disciplines, by contrast, have relatively less rigid theoretical foundations, are more subject to trends or interpretations, and grapple with less clearly defined problems. Pure disciplines are self-regulating, primarily driven by internal standards and academic rigor rather than external demands; and their main objective is the pursuit of knowledge for its own sake rather than its immediate application to professional or practical issues. Applied disciplines, on the other hand, are driven by external factors such as industry needs, government policies, and societal demands. The research and theories developed are geared towards tackling real-world problems and finding direct relevance in various professions. Under this four-cell matrix, disciplinary communities can be categorised into four intellectual groupings, i.e., hard-pure, soft-pure, hard-applied, and soft-applied, which are typified by pure sciences (e.g., Physics), humanities (e.g., History) and pure social sciences (e.g., Anthropology), technologies (e.g., Mechanical Engineering), and applied social sciences (e.g., Education).

The broad grouping of hard *vs* soft and pure *vs* applied disciplines has been widely adopted in current cross-disciplinary investigations (e.g., Hu & Liu, 2018; Hyland, 2000), which set out to explore and identify the dynamics of knowledge-making and academic identity in different domains as reflected via their language choices (Liu et al., 2023). As Becher and Trowler (2001) emphasise, fundamental distinctions between disciplines often emerge through the medium of language, so that a detailed analysis of disciplinary discourses can reveal the characteristics of disciplinary cultures and highlight various facets of their associated knowledge domains. In other words, the distinctive epistemologies of disciplines are encoded in linguistic forms and rhetorical features, thus making linguistic analysis instrumental in unravelling disciplinarity and disciplinary variations (see also Becher, 1987).

To date, cross-disciplinary research can be broadly divided into three main strands. The first strand focuses on the rhetorical move structures in various written or spoken academic genres. For instance, Hyland (2000) examined the move structures of 800 R A abstracts across eight disciplines (Philosophy, Sociology, Applied Linguistics, Marketing, Electrical Engineering, Mechanical Engineering, Physics, and Biology). His investigation showed that while some common rhetorical moves were identified, there were considerable variations in the sequencing of these moves across disciplines. Specifically, writers in the soft knowledge domains preferred to situate their abstract with an Introduction move while writers in the hard knowledge fields tended to omit this move and foreground the Method. The differences observed suggested that academic writers establish credibility, construct research significance, and apply persuasion in RA abstracts in discipline-specific manners. Hu and Liu (2018) reported another cross-disciplinary study on the rhetorical structure of the 3 MT presentation by PhD students from four different disciplines – Biological Sciences, Mechanical Engineering, History, and Education, which were chose to operationalise the hard *vs* soft and pure *vs* applied disciplinary distinctions. Their analysis revealed that the hard-discipline presenters were more likely to employ the Methods move and used less the Framework move than their soft-discipline counterparts; meanwhile, the pure-discipline presenters were more inclined to employ a Results move than their applied-discipline counterparts. The differences were then explained by the dominant epistemological orientations and knowledge-making practices in the respective disciplines.

The second strand of research examines how disciplinary discourses differ in a range of lexico-grammatical or phraseological features. For instance, drawing on the interpersonal model of metadiscourse, Hyland and Jiang and Hyland (2020) examined the diachronic changes in the use of stance markers in disciplinary academic prose and observed a marked decline in hedges, boosters, and attitude markers particularly in the soft fields (e.g., Applied Linguistics and Sociology). In a similar vein, Hyland and Jiang and Hyland (2020) explored how authors in four disciplines rhetorically 'sell' or 'hype' their studies over three time periods. Their study noted more marked increases of hyping in the hard sciences which manifested the hard-discipline authors' greater willingness to display positive attitude, underline their contribution, and hype research primacy, methods and the author's prior research. Concerning the use of lexical bundle, Cortes (2004) compared published writing between History and Biology, focusing on both their structures and functions, which revealed that lexical bundles in Biology covered a wider range of structures and were more likely used to hedge the authors' arguments. In addition, Hyland and Jiang and Hyland (2020) explored changes in the use of lexical bundles in research articles across four disciplines (i.e., Applied Linguistics, Sociology, Electrical Engineering, and Biology) over the past 50 years, and found that pure sciences significantly reduced the number of lexical bundles they employed, while it was the other way around for applied disciplines. They also reported that the hard science fields exhibited a major shift away from verb-phrase bundles towards noun/prepositional types. From a syntactic complexity perspective, Lu et al. (2021) investigated the variation in the relationship between syntactic complexity and rhetorical move-steps in RA introductions in two pure science disciplines (Anthropology and Sociology) and two applied disciplines (Chemical Engineering and Electrical Engineering). The results revealed both similarities and significant differences in terms of how disciplinary writers engaged with syntactic complexity in realising each of the six rhetorical move-steps commonly found in RA introductions, offering an enriched understanding of disciplinary variation in form-function mappings.

The last strand is situated within the Systemic Functional Linguistics, particularly the Appraisal framework (Martin & White, 2005), which investigates how interpersonal meaning is crafted in disciplinary discourses. For example, Zolfaghari (2023) examined the appraisal resources co-articulated with negation in RAs in six disciplines (Astrophysics, Chemistry, Geology, Psychology, Linguistics, and Political Science). The results revealed a significantly greater overall use of appraisal resources in soft disciplines and, specifically, a higher prevalence of heteroglossic engagement, attitude, and graduation devices in these disciplines. Conversely, the hard sciences demonstrated a more concentrated use of appreciation, neutral polarity of attitude, and sharpening resources. Recent advancement in this line of inquiry is further exemplified by Deng et al. (2024) who, drawing on the Engagement subsystem of the Appraisal framework, explored how researchers in four disciplines (Applied Linguistics, Education, Electrical Engineering, and Biology) engage with their discourse community in different move-steps in RA introductions. The study identified, for one thing, clear form-function mappings between the move-steps and the evaluative language resources, and for the other, both commonalities and variations on engagement values between the natural and social sciences.

To recapitulate, the studies reviewed above provide a compelling illustration of the considerable variation of language use across disciplines in terms of rhetorical structures, lexico-phraseological features, and interpersonal meanings. These studies have shed important light into EAP writing pedagogy, inspiring EAP practitioners to offer instructions on discipline-specific writing conventions that are tailored to students' needs, rather than simply teaching them the general principles of academic writing. However, despite the sustained scholarly interest in disciplinary variations in academic discourses, previous research has primarily focused on the generic or linguistic variations at the hard/soft dimension, with relatively fewer efforts to compare the pure and applied disciplines. Furthermore, although the extant literature has documented disciplinary variations at the macro structural level, and their lexico-phraseological aspects, less attention has been paid to writers' pragmatic performance of discourse acts or functions in disciplinary discourses. Therefore, it is pertinent to seek an appropriate approach to explore how writers in different knowledge domains (i.e., hard *vs* soft and pure *vs* applied) perform discourse acts or functions in discipline-specific ways. To this end, we draw on insights from the analytic framework of local grammar, which will be introduced in the following section.

2.2. Local grammar approach to contrastive academic discourse studies

Situated in corpus linguistics, local grammar has been considered an alternative approach to the description and theorising of language use, with each local grammar seeking to account for one discourse act or function only (Hunston & Su, 2019). It differs from general grammar in that it uses functionally transparent and context-specific terminologies, rather than traditional grammatical terms

such as Subject and Object, to analyse corresponding syntactic components of instances realising a particular discourse act or function (Hunston & Sinclair, 2000; Su & Wei, 2018). To illustrate, Table 1 presents the general grammar analysis and local grammar analysis of the sentence *We define absolute frequency as the raw occurrence of linguistic items in the absence of context.*[1]

Table 1 illustrates that, while general grammar could only reveal the syntactic relationship between the elements, local grammar goes further by not only capturing the pragmatic function of the entire sentence but also elucidating the functional role of each syntactic component (for example, *absolute frequency* functions as the item being defined and is thus labelled 'Definiendum'). Therefore, local grammars contribute to a more transparent and adequate description of language use.

Indeed, several studies have employed the local grammar approach to account for discourse acts that are crucial to academic writing. For instance, Allen (2005) developed the local grammar of cause and effect in Biomedical research articles. Similarly, two other studies established the local grammars of exemplification and definition respectively in Linguistics research articles (Su & Zhang, 2020; Zhang & Su, 2022). More recently, Zhang et al. (2024) explored the realisational patterns of graphic data commentary in Business research articles.

What is particularly relevant to the current inquiry is the advantage of local grammars in quantifying the realisations of a particular discourse act at both the lexico-grammatical and discourse-semantic levels. As Zhang and Su (2023) pointed out, instances of a given discourse act co-occurring with different lexical markers may realise the same local grammar pattern, whereas the same marker may trigger distinct patterns. Therefore, rather than simply counting the frequency of lexical markers, local grammar enables the calculation of the discourse-semantic patternings of these linguistic items. This, in turn, could facilitate contrastive academic discourse studies that probe into the variations in the performance of discourse acts among different writer groups or across different contexts. For instance, Su et al. (2021) adopted the local grammar approach to trace the changing patterns of exemplification in Linguistics research articles across two 15-year time periods. Their investigation revealed that, although the frequency of exemplificatory markers remained relatively stable over time, that of the local grammar patterns associated with exemplification underwent significant changes. Subsequent studies further explored the divergences between Chinese English-major MA students and expert writers in their use of exemplification and definition in academic writing (Su et al., 2022; Zhang & Su, 2023). Collectively, the two studies found that the Chinese EFL learners used substantially more definitions but fewer exemplifications in their MA theses compared to expert writers in research articles, with more nuanced similarities and differences observed between the two cohorts.

Synthesising the aforementioned studies underscores the usefulness of local grammars in contrastive academic discourse studies by uncovering the underlying regularity of language use at a more abstract level, i.e., the discourse-semantic level. However, the existing local grammar research on academic discourse acts has been restricted to single disciplines (e.g., Biomedicine, Linguistics, or Business). Furthermore, given that previous studies investigating disciplinarity have primarily focused on macro-structural or lexico-grammatical features, it is imperative to further examine how discourse-semantic features vary across disciplines. Bringing these together points to the necessity and possibility of extending the local grammar approach to cross-disciplinary research, which will be explored in the present study.

3. Methodology

3.1. Corpora

The data for the current investigation were drawn from the DEAP (Database of English for Academic Purposes) Corpus, which can be freely accessed via BFSU CQPweb (http://114.251.154.212/cqp/). The DEAP Corpus consists of a series of sub-corpora which comprise English research articles published in high-impact international journals from more than 20 disciplines. Essentially, each DEAP sub-corpus is specialised in one discipline, thus providing rich sources for exploring the unique linguistic characteristics and communication styles within each discipline. We selected four sub-corpora from the DEAP Corpus, i.e., LinDEAP (Linguistics), EduDEAP (Education), PhysDEAP (Physics), and CivDEAP (Civil Engineering), which, according to Becher and Trowler (2001), operationalise the hard *vs*. soft and pure *vs*. applied disciplinary dichotomies (as illustrated in Fig. 1). The four disciplines were selected because they are considered established representative disciplines in the respective intellectual clusters (Becher, 1987) and have often been examined in cross-disciplinary discourse studies (e.g., Hu & Liu, 2018; Hyland, 2000). The meta-information of the four sub-corpora is given in Table 2.

At this juncture, we should note that selecting the four disciplines to represent the hard/soft and pure/applied dichotomies may raise a potential issue. That is, in so doing we have neglected the internal variability across the sub-disciplinary areas. For example, within Linguistics (which is labelled soft-pure), there are strands (e.g., computational linguistics, natural language processing) that are 'harder' than the mainstream of the discipline and other strands (e.g., applied linguistics) that are more concerned with translating

[1] Unless otherwise indicated, all examples used in this paper are taken from the corpora used in this study.

Table 1
General grammar analysis *vs*. local grammar analysis.

General grammar	Subject	Predicate	Object	Complement
	We	*define*	*absolute frequency*	*as the raw occurrence of linguistic items in the absence of context.*
Local grammar	Source-internal[1]	Hinge	Definiendum	Definiens

[1] The explanation of the terminologies used in local grammar analyses of definition will be introduced in Section 3.3.

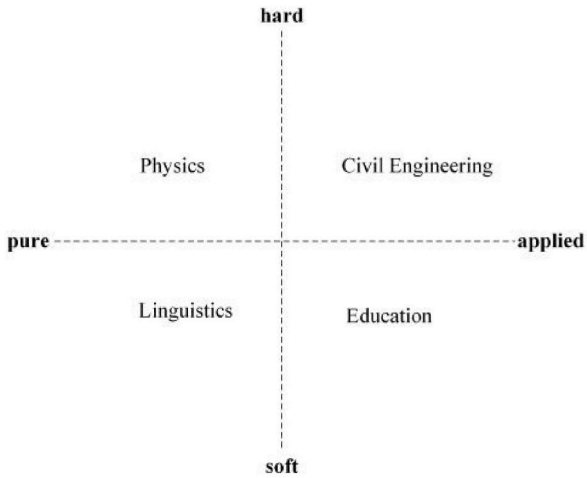

Fig. 1. Knowledge domains and representative disciplines (adapted from Groom, 2007, p. 17).

Table 2
Metainformation of the four sub-corpora.

Sub-corpora	Number of tokens	Number of texts	Average length
LinDEAP	5,849,963	626	9344.99
EduDEAP	5,688,598	673	8452.60
PhysDEAP	4,838,844	487	9936.02
CivDEAP	5,731,292	780	7347.81
Total	**22,108,697**	**2566**	**8770.36**

linguistic theories into the practical enhancement of language teaching and learning. This intradisciplinary diversity has been acknowledged by Hu and Wang (2014) who suggested that a solution for this problem lies in constructing a representative sample of RAs for each discipline. Since the DEAP Corpus is a hybrid corpus sampling across a broad spectrum of subdisciplines for each sub-corpus,[2] it is arguable that the four selected sub-corpora are largely representative of the general landscape of the parent disciplines.

3.2. Data retrieval

Following previous studies (e.g., Flowerdew, 1992; Triki, 2019; Zhang & Su, 2022, 2023), we identified a set of lexical items and one syntactic construction signalling the discourse act of definition. For the sake of clarity, these markers were grouped into four broad categories (see also Zhang & Su, 2022, 2023): definitional construction, definitional nouns, definitional verbs, and definitional linking adverbials. We then searched for these items and their possible variants in the four corpora. The quantitative information of these markers and their distribution across the four corpora are presented in Table 3.

[2] For instance, LinDEAP comprises RAs which were roughly equally sampled from 10 subfields of Linguistics: applied linguistics, cognitive linguistics, corpus linguistics, discourse analysis and stylistics, natural language processing (NLP), psycholinguistics, pragmatics, semantics and lexicography, sociolinguistics, and theoretical linguistics. Although there could be debate regarding the classification of computational linguistics and NLP as distinct fields of Linguistics, as suggested by one reviewer, they are generally regarded as subfields of Linguistics in the context of higher education in China where the DEAP Corpus originated. We, however, agree with the reviewer that the inclusion of such fields under the broader umbrella of Linguistics may be potentially controversial and would therefore suggest that future studies carefully consider this aspect when design their corpora.

Regarding data retrieval, a few caveats need to be noted. First, searching for these markers may return hits that do not actually fulfil the function of defining. A case in point is the search of the 'X BE Y' construction. In the DEAP Corpus, the script for searching for this construction is "_NN* _VB* (_AT*)? * _NN*", allowing up to two words that may occur between the copula 'BE' and 'Y'. The search, however, may return many hits that are descriptive rather than definitive (e.g., *Assessment of clinical competence post-training is an important aspect of program evaluation.*). Such instances were excluded at the later stage of local grammar analysis. Second, instances of definition that contain multiple markers listed in Table 3 (e.g., *The term stuttering severity refers to the level of disruption in the delivery of continuous …*) were analysed and counted once to avoid repetition (see also Zhang & Su, 2022). Finally, it is important to mention that using a computerised search we could only capture those overt definitions that are explicitly signalled by the predetermined lexico-syntactic markers. In other words, we have not considered those covert ones that lack such clear signals, and we acknowledge this as a limitation of the current investigation.

3.3. Data analysis

As previously mentioned, local grammar uses context-specific and transparent functional terminologies to analyse the corresponding syntactic components of instances associated with an intended discourse act. Thus, the identification of a set of terminologies is crucial for the subsequent local grammar analyses of definition. In this study we adapted the terminologies used in Zhang and Su (2022), which were reproduced and explained succinctly in Table 4.

The two authors of the current investigation independently performed the local grammar analyses of all instances retrieved from the corpora. Debriefing sessions were held regularly to discuss borderline cases and disagreements between the two coders, which to a large extent ensured the reliability of the data analyses. Log-likelihood tests were conducted using Log-likelihood and Chi-square Calculator (Version 1.0) (available at https://corpus.bfsu.edu.cn/TOOLS.htm) to examine whether the variations in the distributions of local grammar patterns of definition between the soft and hard disciplines, and between the pure and applied disciplines, are statistically significant. In the following section we present and discuss the results.

4. Results and discussion

4.1. Local grammar patterns of definition in disciplinary discourses

Our analyses identified 37 local grammar patterns of definition occurring more than one time per million words in each of the four corpora. Meanwhile, there were also quite a number of distinct yet fairly infrequent patterns in the four disciplines, i.e., less than one time per million words in all four corpora. These infrequent patterns were considered idiosyncratic and excluded from the subsequent discussion. Following Zhang and Su (2022, 2023), we categorised the 37 local grammar patterns into five sets according to their prototypicality and similarity. Specifically, the first set includes five patterns consisting of the four fundamental elements of definition, i.e., 'Definiendum', 'Hinge', 'Definiens', and 'Supporting statement'. Patterns in the other four sets can be regarded as variants of those included in Set 1, in which a definition is made more explicit through additional elements. Set 2 contains twenty patterns in two subtypes, one attributing a definition to previous literature or other researchers, labelled 'Source-external', and the other to the authors themselves, labelled 'Source-internal'. The third set of analyses includes seven patterns in which the writers delimit the extent to which

Table 3
Markers of definition and their distribution in the four corpora.

		LinDEAP		EduDEAP		PhysDEAP		CivDEAP	
	Markers	Raw	Normed[1]	Raw	Normed	Raw	Normed	Raw	Normed
Definitional construction	'X BE Y' construction	9405	1607.7	7923	1392.79	6786	1402.4	8023	1399.86
Definitional nouns	term (as noun)	1136	194.19	864	151.88	2591	535.46	542	94.57
	concept	909	155.39	1128	198.29	349	71.12	608	106.08
	definition	759	129.74	547	96.16	878	181.45	512	89.33
	notion	905	154.7	435	76.47	168	34.72	43	7.5
Definitional verbs	consider	3650	623.94	3762	661.32	4266	881.62	6352	1108.3
	refer	1659	283.59	1095	192.49	482	99.61	713	124.4
	define	1895	323.93	1548	272.12	3127	646.23	2844	496.22
	consist of	932	159.32	721	126.74	760	157.06	1179	205.71
	mean (as verb)	2088	356.93	1514	266.15	856	176.9	1104	192.63
	seen as	415	70.94	348	61.18	118	24.39	55	9.6
	comprise	383	65.47	461	81.04	176	36.37	354	61.77
	call (as verb)	1580	270.09	1024	180.01	1088	224.85	498	86.89
	known as	175	29.91	123	21.62	333	68.82	213	37.16
	regard (as verb)	310	52.99	428	75.24	213	44.02	394	68.75
	term (as verb)	179	30.6	89	15.65	94	19.43	83	14.48
Definitional linking adverbials	i.e.	2962	506.33	2457	431.92	2205	455.69	3585	625.51
	that is	3508	599.66	1737	305.35	1651	341.2	1051	183.38
	BE related to	333	56.92	590	103.72	410	84.73	436	76.07
	BE concerned with	115	19.66	54	9.49	36	7.44	19	3.32

[1] The data were normalised per million words, which applies throughout the study.

Table 4
Terminologies for local grammar analysis of definition (adapted from Zhang & Su, 2022).

Terminology	Explanation	Example
Definiendum	The term or concept being defined.	***Positioning*** *is defined by Du Bois as "the act of situating a …*
Definiens	The element that indicates how the term or concept is defined.	Recurrent miscarriages are defined as ***three or more consecutive losses*** …
Source-internal	The term or concept is being defined by the author(s) him/herself (/themselves).	***We*** define funniness of a text as a numerical value …
Source-external	The term or concept is being defined by anyone else other than the author(s).	***Many authors*** refer to the intersective theory as Boolean conjunction …
Hinge	The element that links different parts of a definition.	VFS ***can be seen as*** a claim that the logic …
Initiator	The element that indicates that a definition is being given.	***The term*** "interaction" is understood here as the likelihood with which a …
Scope	The element that indicates the degree to which a definition is valid.	***In the present study,*** "speech disruption" was defined for participants as "the level …
Supporting statement	The element that provides additional information for how a term/ concept is being defined.	"Unpassive" is a biased term ***in that it suggests the pattern's inclusion in the*** …

a given definition is valid or applicable through an element 'Scope'. Set 4 includes two patterns in which the definition is introduced by an 'Initiator', which is usually marked by definitional nouns, such as *concept*, *definition*, *notion*, etc. The last set contains three patterns that are structurally more complex, comprising two or more additional elements. Table 5 provides an overview of the 37 local grammar patterns in these five sets, with an example and quantitative information given for each.

Overall, data in Table 5 reveal that across the four disciplines, patterns in Set 1 are used with the highest frequency, occurring 344.43 times per million words, while patterns in Set 5 are employed the least, appearing only 3.30 times per million words. This disparity shows that the disciplinary authors have a marked preference for the conventional patterns when crafting definitions and seldom resort to the intricate ones.

Table 6 displays the distribution of the five sets of local grammar patterns of definition across the four disciplines, from which several observations can be tentatively drawn. First, definitions appear most frequently in Education, with 505.92 instances per million words. This may suggest that writers in Education are particularly inclined to leverage definitions to elaborate key educational concepts or terms. Conversely, definitions are least prevalent in Linguistics, occurring 410.26 times per million words, which means that linguists are relatively less likely to elucidate concepts or terms through the act of defining than their counterparts in the other three disciplines. One possible explanation is that language is inherently complex and linguistic theories and models evolve over time, with many aspects open to interpretation; therefore, Linguistics deals with many abstract concepts (like phonology, syntax, and semantics), which require extensive explanation rather than simple, universally agreed-upon definitions. By contrast, educational research often involves defining terms and concepts to create a framework for studying learning processes, teaching methods, and educational outcomes. In the field of Education, there is a need for standardised language and understanding across different regions, institutions, and educational systems. Clear definitions help ensure that key terms and concepts are easy to understand, which in turn fosters the collective understanding and advancement of educational practices worldwide.

4.2. Variations in the distribution of local grammar patterns of definition between the soft and hard disciplines

Table 7 presents the distribution of local grammar patterns of definition between the soft and hard disciplines. In general, definitions are slightly more prevalent in hard disciplines than in soft disciplines, although this difference lacks statistical significance (LL $= -1.83$, $p = 0.1764$). However, a detailed examination into each set reveals more pronounced similarities and differences between the two domains.

Set 1 is markedly more frequent in hard disciplines than in soft disciplines (LL $= -133.71$, $p < 0.0001$). In particular, the higher frequency of PA1, PA3, and PA4 in hard disciplines indicates a stronger reliance on the simpler and more prototypical patterns by hard-discipline writers. Specifically, PA1 is the simplest pattern wherein 'Definiendum' is positioned in the left side of the definition clause and 'Definiens' on the right side, connected by a 'Hinge' (see also Zhang & Su, 2022). In contrast, the way in which definition is performed in PA3 and PA4 features the reversed order, i.e., 'Definiens' precedes 'Definiendum' in a definition sentence. In Systemic Functional Linguistics (Halliday & Matthiessen, 2014), these two patterns place the 'Definiens' as the 'Theme' (or the *Given* information) and 'Definiendum' as the 'Rheme' (or the *New* information). This means that, arguably, hard discipline writers show a higher tendency to enact their definitions by naming the 'Definiens', compared to their counterparts in soft disciplines. Table 8 provides an example of PA3 from CivDEAP for further illustration.

Patterns in Set 2 are more than twice as frequent in soft disciplines as in hard disciplines (LL $= 407.39$; $p < 0.0001$). Out of the 20 patterns in this set, 16 patterns occur more frequently in soft disciplines. This indicates that writers in soft disciplines are more inclined to specify the source of their definitions. More notably, only three out of the 16 patterns – PA8, PA19, and PA25 – are characterised by 'Source-internal', while the remaining 13 patterns implement definitions derived from external sources. This finding clearly shows that writers in soft disciplines are more likely to draw on externally sourced definitions compared to the hard discipline writers, which may be attributed to the reiterative nature of knowledge construction in soft disciplines, where scholars frequently revisit and reinterpret existing materials before formulating opinions in relation to others' views (Becher & Trowler, 2001; Charles, 2006).

While there was no significant variance concerning the overall use of patterns in Set 3, PA27 and PA30 exhibit statistically

Table 5
An overview of the local grammar patterns of definition in the four corpora.

Set	No.	Patterns	Raw	Normed
1	PA1	Definiendum – Hinge – Definiens e.g., Demotions are moves downward from a principal position in year to a school-based, non-principal position in year t+1.	3586	162.20
	PA2	Definiendum – Hinge – Definiens – Supporting statement e.g., Thus, the community is also defined as a place where information is reliable and trustworthy.	2278	103.04
	PA3	Definiens – Hinge – Definiendum e.g., Thus, this composite was called as shape stabilized-composite PCMs (SS-CPCMs).	1400	63.32
	PA4	Definiens – Hinge – Definiendum – Supporting statement e.g., This method is called the replica trick that is often employed for the entanglement …	206	9.32
	PA5	Definiens – Supporting statement – Hinge – Definiendum e.g., This artificial path that does not correspond to any possible contour is refereed next as the virtual IBZ contour.	145	6.56
		Subtotal	**7615**	**344.43**
2	PA6	Source-external – Hinge – Definiendum – Definiens e.g., Hahs-Vaughn et al. (2017) defined statistical literacy as an understanding of fundamental statistical concepts.	247	11.17
	PA7	Definiendum – Hinge – Definiens – Source-external e.g., Social support can be defined as a multidimensional concept (Dumont & Provost, 1999) …	245	11.08
	PA8	Source-internal – Hinge – Definiens – Definiendum e.g., We refer to this protocol as HSE06.	208	9.41
	PA9	Definiens – Hinge – Definiendum – Source-external e.g., This level of professional knowledge can be defined as an orientation scheme (Bohnsack, 2017a).	185	8.37
	PA10	Source-internal – Hinge – Definiendum – Definiens e.g., The authors defined semantic signatures as a set of highly related and interlinked WordNet senses.	163	7.37
	PA11	Source-external – Hinge – Definiendum – Definiens – Supporting statement e.g., Bateson and Goffman have defined a frame as the definition that individuals attribute to a …	133	6.02
	PA12	Definiendum – Hinge – Definiens – Supporting statement – Source-external e.g., An establishment is an economic unit that produces goods or provides services (BLS 2017).	105	4.75
	PA13	Definiens – Source-internal – Hinge – Definiendum e.g., …longitudinal stress of a piezoelectric control device, which we call the piezoelectric transducer.	83	3.75
	PA14	Source-internal – Hinge – Definiendum – Definiens – Supporting statement e.g., We define the onset time as the time at which the standard …	73	3.30
	PA15	Source-external – Hinge – Definiens – Definiendum e.g., Darabi et al. (2012) called this variable the effective damage variable …	71	3.21
	PA16	Source-external – Hinge – Definiendum – Hinge – Definiens e.g., Wenger (1991) called a "learning curriculum", defined as "situated opportunities … for the improvisational development of …	69	3.12
	PA17	Source-external – Definiendum – Hinge – Definiens e.g., According to the eigenforce model, the interaction energy is considered as the integral of induced eigenforces times displacement over each …	61	2.76
	PA18	Definiens – Source-external – Hinge – Definiendum e.g., …pausing, and tone-features that Gumperz (1982) calls contextualization cues.	55	2.49
	PA19	Source-internal – Hinge – Definiendum – Hinge – Definiens e.g., I also proposed that SG and WG consist of core and peripheral grammatical elements.	53	2.40
	PA20	Source-external – Definiendum – Hinge – Definiens – Supporting statement e.g., According to Chomsky (1965), selectional restrictions are lexicon-internal constraints that verbs place on their arguments; …	38	1.72
	PA21	Definiendum – Hinge – Source-external – Definiens e.g., Contrastingly, improvisation was considered by the student-participants to be a shared, time-bound activity …	33	1.49
	PA22	Definiendum – Source-external – Hinge – Definiens e.g., …technacy framework by Seeman and Talbot (1995) is defined as a holistic approach to perceiving, teaching, practicing …	33	1.49
	PA23	Source-external – Hinge – Definiendum – Hinge – Definiens – Supporting statement e.g., Breban proposes that intersubjectivity should be defined as meanings that are "hearer or addressee centred" …	33	1.49
	PA24	Source-external – Hinge – Definiens – Hinge – Definiendum – Supporting statement e.g., Liverier subsequently proposed a promising scheme called multidimensional reconciliation, which is able to build a virtual …	26	1.18
	PA25	Source-internal – Hinge – Definiendum – Hinge – Definiens – Supporting statement e.g., …we use the term constituent to refer to lexical items that make up MWEs.	22	1.00
		Subtotal	**1936**	**87.57**
3	PA26	Scope – Definiendum – Hinge – Definiens e.g., In this example, a pure product state is then considered as a state carrying "one unit of quantumness".	167	7.55
	PA27	Scope – Definiendum – Hinge – Definiens – Supporting statement e.g., In this study, the apparent cohesion is a function of suction stress, which varies with depth.	99	4.48
	PA28	Definiendum – Scope – Hinge – Definiens e.g., Epistemology here refers to the study of knowledge and in particular the investigation.	58	2.62
	PA29	Definiens – Hinge – Definiendum – Scope e.g., This alternative view of grammar is often called usage-based or cognitive-functional in the literature.	36	1.63
	PA30	Scope – Definiens – Hinge – Definiendum e.g., In the literature this method is often called the Ritus method.	26	1.18

(*continued on next page*)

Table 5 (*continued*)

Set	No.	Patterns	Raw	Normed
	PA31	Definiendum – Scope – Hinge – Definiens – Supporting statement e.g., …an HEI in a developing country is comprised of faculties, which houses academic departments …	25	1.13
	PA32	Definiendum – Hinge – Scope – Definiens e.g., PP is defined in the present paper as remembering, comparing, and …	22	1.00
		Subtotal	**433**	**19.59**
4	PA33	Initiator – Definiendum – Hinge – Definiens e.g., The notion of transfer function refers to the ratio between the Laplace transform of a certain output entity …	119	5.38
	PA34	Initiator – Definiendum – Hinge – Definiens – Supporting statement e.g., The concept of resilience refers to findings that some individuals have relatively good psychological outcomes, despite exposure to …	68	3.08
		Subtotal	**187**	**8.46**
5	PA35	Initiator – Definiendum – Source–internal – Hinge – Definiens e.g., By elite schools, we mean public schools with selective merit-based admission.	25	1.13
	PA36	Initiator – Definiendum – Source-internal – Hinge – Definiens – Supporting statement e.g., By at-issue, we mean content that expresses the main point or main assertion of …	25	1.13
	PA37	Scope – Source-internal – Hinge – Definiendum – Definiens e.g., In the current framework, we define ext as the mean at the class level …	23	1.04
		Subtotal	**73**	**3.30**
		Total	**10244**	**463.35**

Table 6
Distribution of local grammar patterns of definition across the four corpora.

Set	Pattern	LinDEAP			EduDEAP			PhysDEAP			CivDEAP		
		Raw	Normed	per text	Raw	Normed	per text	Raw	Normed	per text	Raw	Normed	per text
1	PA1	754	128.89	1.20	904	158.91	1.34	682	140.94	1.40	1246	217.40	1.60
	PA2	552	94.36	0.88	717	126.04	1.07	419	86.59	0.86	590	102.94	0.76
	PA3	140	23.93	0.22	294	51.68	0.44	612	126.48	1.26	354	61.77	0.45
	PA4	11	1.88	0.02	21	3.69	0.03	98	20.25	0.20	76	13.26	0.10
	PA5	25	4.27	0.04	52	9.14	0.08	47	9.71	0.10	21	3.66	0.03
	Subtotal	**1482**	**253.33**	2.37	**1988**	**349.47**	2.95	**1858**	**383.98**	3.82	**2287**	**399.04**	2.93
2	PA6	70	11.97	0.11	146	25.67	0.22	14	2.89	0.03	17	2.97	0.02
	PA7	82	14.02	0.13	97	17.05	0.14	16	3.31	0.03	50	8.72	0.06
	PA8	116	19.83	0.19	33	5.80	0.05	50	10.33	0.10	9	1.57	0.01
	PA9	62	10.60	0.10	48	8.44	0.07	43	8.89	0.09	32	5.58	0.04
	PA10	49	8.38	0.08	46	8.09	0.07	61	12.61	0.13	7	1.22	0.01
	PA11	53	9.06	0.08	60	10.55	0.09	7	1.45	0.01	13	2.27	0.02
	PA12	23	3.93	0.04	57	10.02	0.08	5	1.03	0.01	20	3.49	0.03
	PA13	38	6.50	0.06	22	3.87	0.03	18	3.72	0.04	5	0.87	0.01
	PA14	29	4.96	0.05	15	2.64	0.02	24	4.96	0.05	5	0.87	0.01
	PA15	33	5.64	0.05	26	4.57	0.04	7	1.45	0.01	5	0.87	0.01
	PA16	34	5.81	0.05	22	3.87	0.03	5	1.03	0.01	8	1.40	0.01
	PA17	33	5.64	0.05	20	3.52	0.03	1	0.21	0.00	7	1.22	0.01
	PA18	24	4.10	0.04	25	4.39	0.04	3	0.62	0.01	3	0.52	0.00
	PA19	27	4.62	0.04	11	1.93	0.02	8	1.65	0.02	7	1.22	0.01
	PA20	16	2.74	0.03	21	3.69	0.03	1	0.21	0.00	0	0.00	0.00
	PA21	9	1.54	0.01	13	2.29	0.02	0	0.00	0.00	11	1.92	0.01
	PA22	18	3.08	0.03	11	1.93	0.02	1	0.21	0.00	3	0.52	0.00
	PA23	23	3.93	0.04	7	1.23	0.01	1	0.21	0.00	2	0.35	0.00
	PA24	1	0.17	0.00	3	0.53	0.00	8	1.65	0.02	14	2.44	0.02
	PA25	12	2.05	0.02	8	1.41	0.01	1	0.21	0.00	1	0.17	0.00
	Subtotal	**752**	**128.55**	1.20	**691**	**121.47**	1.03	**274**	**56.63**	0.56	**219**	**38.21**	0.28
3	PA26	36	6.15	0.06	41	7.21	0.06	45	9.30	0.09	45	7.85	0.06
	PA27	25	4.27	0.04	40	7.03	0.06	9	1.86	0.02	25	4.36	0.03
	PA28	3	0.51	0.00	28	4.92	0.04	20	4.13	0.04	7	1.22	0.01
	PA29	6	1.03	0.01	8	1.41	0.01	3	0.62	0.01	19	3.32	0.02
	PA30	2	0.34	0.00	3	0.53	0.00	11	2.27	0.02	10	1.74	0.01
	PA31	4	0.68	0.01	10	1.76	0.01	5	1.03	0.01	6	1.05	0.01
	PA32	7	1.20	0.01	5	0.88	0.01	0	0.00	0.00	10	1.74	0.01
	Subtotal	**83**	**14.19**	0.13	**135**	**23.73**	0.20	**93**	**19.22**	0.19	**122**	**21.29**	0.16
4	PA33	41	7.01	0.07	27	4.75	0.04	29	5.99	0.06	22	3.84	0.03
	PA34	21	3.59	0.03	21	3.69	0.03	9	1.86	0.02	17	2.97	0.02
	Subtotal	**62**	**10.60**	0.10	**48**	**8.44**	0.07	**38**	**7.85**	0.08	**39**	**6.80**	0.05
5	PA35	11	1.88	0.02	3	0.53	0.00	11	2.27	0.02	0	0.00	0.00
	PA36	8	1.37	0.01	8	1.41	0.01	9	1.86	0.02	0	0.00	0.00
	PA37	2	0.34	0.00	5	0.88	0.01	13	2.69	0.03	3	0.52	0.00
	Subtotal	**21**	**3.59**	0.03	**16**	**2.81**	0.02	**33**	**6.82**	0.07	**3**	**0.52**	0.00
	Total	**2400**	**410.26**	3.83	**2878**	**505.92**	4.28	**2296**	**474.49**	4.71	**2670**	**465.86**	3.42

Table 7
Distribution of local grammar patterns of definition in soft and hard disciplines.

Set	Pattern	Soft (LinDEAP/EduDEAP)		Hard (PhysDEAP/CivDEAP)		LL	Sig
		Raw	Normed	Raw	Normed		
1	PA1	1658	143.69	1928	182.40	−50.90	****0.0000
	PA2	1269	109.98	1009	95.46	+11.32	***0.0008
	PA3	434	37.61	966	91.39	−256.66	****0.0000
	PA4	32	2.77	174	16.46	−120.49	****0.0000
	PA5	77	6.67	68	6.43	0.05	0.8257
	Subtotal	3470	300.73	4145	392.14	−133.71	****0.0000
2	PA6	216	18.72	31	2.93	+140.06	****0.0000
	PA7	179	15.51	66	6.24	+44.71	****0.0000
	PA8	149	12.91	59	5.58	+32.77	****0.0000
	PA9	110	9.53	75	7.10	+3.95	*0.0469
	PA10	95	8.23	68	6.43	2.44	0.1183
	PA11	113	9.79	20	1.89	+63.87	****0.0000
	PA12	80	6.93	25	2.37	+25.68	****0.0000
	PA13	60	5.20	23	2.18	+14.00	***0.0002
	PA14	44	3.81	29	2.74	1.93	0.1648
	PA15	59	5.11	12	1.14	+29.93	****0.0000
	PA16	56	4.85	13	1.23	+25.24	****0.0000
	PA17	53	4.59	8	0.76	+33.33	****0.0000
	PA18	49	4.25	6	0.57	+34.68	****0.0000
	PA19	38	3.29	15	1.42	+8.41	** 0.0037
	PA20	37	3.21	1	0.09	+40.35	****0.0000
	PA21	22	1.91	11	1.04	2.84	0.0921
	PA22	29	2.51	4	0.38	+19.24	****0.0000
	PA23	30	2.60	3	0.28	+23.34	****0.0000
	PA24	4	0.35	22	2.08	−15.35	***0.0001
	PA25	20	1.73	2	0.19	+15.56	***0.0001
	Subtotal	1443	125.06	493	46.64	+407.39	****0.0000
3	PA26	77	6.67	90	8.51	−2.47	0.1158
	PA27	65	5.63	34	3.22	+7.34	**0.0067
	PA28	31	2.69	27	2.55	0.04	0.8478
	PA29	14	1.21	22	2.08	−2.56	0.1094
	PA30	5	0.43	21	1.99	−12.04	***0.0005
	PA31	14	1.21	11	1.04	0.15	0.7025
	PA32	12	1.04	10	0.95	0.05	0.8248
	Subtotal	218	18.89	215	20.34	−0.59	0.4427
4	PA33	68	5.89	51	4.82	1.18	0.2783
	PA34	42	3.64	26	2.46	2.53	0.1118
	Subtotal	110	9.53	77	7.28	3.32	0.0684
5	PA35	14	1.21	11	1.04	0.15	0.7025
	PA36	16	1.39	9	0.85	1.42	0.2333
	PA37	7	0.61	16	1.51	−4.45	*0.0349
	Subtotal	37	3.21	36	3.41	−0.07	0.7969
	Total	5278	457.42	4966	469.81	−1.83	0.1764

Note: LL = log-likelihood value; Sig = significance; '+' or '–' = frequency being significantly higher or lower in soft relative to hard disciplines; * = $p < 0.05$; ** = $p < 0.01$; *** = $p < 0.001$; **** = $p < 0.0001$.

Table 8
Example of PA3 in CivDEAP.

Definiens	Hinge	Definiendum
The points distributed on the boundary	*are called*	*surface Gauss points*

significant differences. Specifically, PA27 is more frequent in soft disciplines than in hard disciplines (LL = 7.34, $p < 0.01$) whereas PA 30 is more common in hard disciplines (LL = −12.04, $p < 0.001$). Since PA27 and PA30 are variants of PA2 and PA3 in Set 1, featuring Scope at the initial position, it can be reasonably speculated that the increased usage of PA 27 in soft disciplines aligns with the higher frequency of PA2 in the same domain; and likewise, the broader application of PA30 in hard disciplines is likely linked to the relatively more frequent occurrences of PA3 in this domain. The remaining five patterns – PA26, PA28, PA29, PA31, and PA32 – are used at a similar frequency in both domains. The distinctiveness of these five patterns lies in the adaptable positioning of 'Scope' within a sentence. This indicates that writers from both domains may flexibly place Scope at the beginning, middle, or end position of a definition sentence. The flexibility to place the Scope at different positions in a definitional sentence allows writers to control emphasis, clarity, and engagement of their utterances, thereby influencing how readers interpret and process the information. For example, an initial placement might suggest that the Scope is central to the definition and immediately grab the reader's attention,

while a delayed placement might alter the flow and structure of the discourse and add nuances to the definition.

With regard to the two patterns in Set 4, it is observed that writers in both soft and hard disciplines employ them with a comparable frequency, showing no statistically significant distinction. This observation echoes Zhang and Su (2023) who also discovered a similar use of these two patterns by both Chinese EFL learner writers and expert writers. It thus can be inferred that local grammar patterns of definition initiated with definitional nouns possess broad applicability across various disciplinary fields and among different groups of writers. Examples of these two patterns are presented in Table 9, which are from LinDEAP and PhysDEAP, respectively.

Patterns in Set 5, which are structurally more complex, are relatively less used by disciplinary writers, with no significant differences observed between the two knowledge domains. However, an exception is PA37, which is significantly more frequent in hard disciplines than in soft disciplines (LL = −4.45, $p < 0.05$). The structure of this pattern is distinguished by the placement of 'Scope' at the initial position followed by 'Source-internal', which are then connected with the core elements of the definition – 'Definiendum' and 'Definiens' – through a 'Hinge'. One probable explanation is that allusions to oneself using first-person pronouns (*I*, *we*, *my*, *me*, *us*, *our*) or terms like '*this author*' would portray the writers as self-assured, asserting their own right to demonstrate disciplinary knowledge and expertise. Furthermore, research in hard disciplines typically involves experimental methods, where abstract notions are often defined within specific experimental settings. Therefore, researchers in these fields may formulate their own definitions within a narrower scope to ensure the objectivity and reproducibility of their research. An example of PA37 from PhysDEAP is given in Table 10.

4.3. Variations in the distribution of local grammar patterns of definition between the pure and applied disciplines

Table 11 presents the distribution of local grammar patterns of definition between pure and applied disciplines. Overall, writers in applied disciplines perform the discourse act of definition at a notably higher rate than those in pure disciplines (LL = −25.78, $p < 0.0001$). According to Becher and Trowler (2001), applied disciplines are market- and user-oriented, thus the knowledge construction in applied disciplines is inherently practical, with research findings often translated into products, technical solutions, or methodologies. Consequently, researchers utilise more defining strategies to present or clarify novel, unclear, or intricate concepts, aiming to suit practical needs and user requests.

The most pronounced difference between the pure and applied fields exists in patterns in Set 1 – PA1 and PA2 in particular, which are considerably more common in applied disciplines compared to pure ones (LL = −61.57, $p < 0.0001$). This indicates that authors in applied disciplines rely more heavily on the prototypical patterns of definition and have utilised them as default methods for defining terms or concepts in their scholarly communication. This may be partly attributed to applied discipline writers' preference for using common defining strategies so as to ensure that outsiders can easily understand the technical terms or products. Additionally, researchers in applied disciplines tend to engage more with craft and technique than with methodology and theory, thus they "… deal in quantifiable and universal entities and value neatness and simplicity of explanation" (Becher, 1987, p. 273).

Patterns in Set 2 are deployed more intensively by pure disciplines (LL = 16.75, $p < 0.0001$), demonstrating that writers in pure disciplines are more inclined to emphasise the source of the definition. This tendency may stem from the nature of pure sciences, which involve theories and concepts that require repeated exploration and validation. In particular, seven patterns – PA8, PA9, PA10, PA13, PA14, PA19, and PA23 – are more frequent in pure disciplines. Notably, five of these seven patterns incorporate 'Source-internal' at various positions in the definitional clauses. By comparison, the other four patterns (PA6, PA7, PA12, and PA21), which attribute the source of a definition to other researchers, are significantly more common in applied disciplines than in pure ones. These findings suggest that writers in pure disciplines place greater emphasis on acknowledging the source of their knowledge, particularly their personal authority in knowledge construction. This is at odds with previous claims that applied disciplines are more concerned with the practicality of knowledge and opt to persuade readers by active author intervention, while the pure sciences emphasise knowledge development through the analysis of literature (Hyland and Jiang, 2018b).

Patterns in Set 3 are used more frequently by authors in applied disciplines than in pure disciplines, particularly PA27 and PA29 which occur more frequently in applied disciplines and show significant differences (LL = −7.93, $p < 0.01$ for PA27; LL = −8.27, $p < 0.01$ for PA29). The heavier use of Scope in applied disciplines compared to pure disciplines could be attributed to the differing priorities of these fields. Applied disciplines emphasise research applicability, set more direct goals, and permit a broader flexibility in interpretation and application. In contrast, pure disciplines are centred around theorisation, maintaining more unified and standardised criteria while attempting to provide in-depth explanations in the pursuit of universal truths. Consequently, writers in applied disciplines are more inclined to hedge their definitions by delimiting the scope of definition, thereby offering a wider interpretative latitude that accommodates varied contexts and alternative perspectives.

Unsurprisingly, there are no discernible differences in the use of patterns in Set 4 between applied and pure sciences. This again reinforces the extensive applicability of definitional nouns across contexts, as previously discussed in Section 4.2.1.

Table 9
Examples of PA33 and PA34.

Initiator	Definiendum	Hinge	Definiens
The traditional notion of	*lemmatisation for spoken languages*	*refers to*	*the practice of …*

Initiator	Definiendum	Hinge	Definiens	Supporting statement
The term	*quark mode*	*refers to*	*a class of models*	*which describe the properties of hadrons …*

Table 10
Example of PA35 from PhysDEAP.

Scope	Source-internal	Hinge	Definiendum	Definiens
In what follows,	*we*	*refer to*	*pk m*	*as the eigenfunction dipole moment …*

Table 11
Distribution of local grammar patterns of definition in pure and applied disciplines.

Set	Pattern	Pure		Applied		LL	Sig
		LinDEAP/PhysDEAP		EduDEAP/CivDEAP			
		Raw	Normed	Raw	Normed		
1	PA1	1436	134.35	2150	188.27	−99.8	****0.0000
	PA2	971	90.84	1307	114.45	−30	****0.0000
	PA3	752	70.35	648	56.74	+16.15	***0.0001
	PA4	109	10.2	97	8.49	1.72	0.1899
	PA5	72	6.74	73	6.39	0.1	0.7526
	Subtotal	3340	312.48	4275	374.35	−61.57	****0.0000
2	PA6	84	7.86	163	14.27	−20.76	****0.0000
	PA7	98	9.17	147	12.87	−6.89	**0.0087
	PA8	166	15.53	42	3.68	+87.51	****0.0000
	PA9	105	9.82	80	7.01	+5.25	*0.0220
	PA10	110	10.29	53	4.64	+24.31	****0.0000
	PA11	60	5.61	73	6.39	−0.56	0.455
	PA12	28	2.62	77	6.74	−20.65	****0.0000
	PA13	56	5.24	27	2.36	+12.36	***0.0004
	PA14	53	4.96	20	1.75	+17.74	****0.0000
	PA15	40	3.74	31	2.71	1.82	0.1777
	PA16	39	3.65	30	2.63	1.85	0.174
	PA17	34	3.18	27	2.36	1.33	0.2479
	PA18	27	2.53	28	2.45	0.01	0.9121
	PA19	35	3.27	18	1.58	+6.73	**0.0095
	PA20	17	1.59	21	1.84	−0.2	0.6557
	PA21	9	0.84	24	2.1	−6.12	*0.0134
	PA22	19	1.78	14	1.23	1.13	0.2883
	PA23	24	2.25	9	0.79	+8.1	**0.0044
	PA24	9	0.84	17	1.49	−2	0.1572
	PA25	13	1.22	9	0.79	1.02	0.3125
	Subtotal	1026	95.99	910	79.69	+16.75	****0.0000
3	PA26	81	7.58	86	7.53	0	0.9677
	PA27	34	3.18	65	5.69	−7.93	**0.0049
	PA28	23	2.15	35	3.06	−1.77	0.1833
	PA29	9	0.84	27	2.36	−8.27	**0.0040
	PA30	13	1.22	13	1.14	0.03	0.8661
	PA31	9	0.84	16	1.4	−1.55	0.213
	PA32	7	0.65	15	1.31	−2.47	0.1159
	Subtotal	176	16.47	257	22.50	−10.36	**0.0013
4	PA33	70	6.55	49	4.29	+5.24	*0.0220
	PA34	30	2.81	38	3.33	−0.49	0.4846
	Subtotal	100	9.36	87	7.62	1.97	0.1605
5	PA35	22	2.06	3	0.26	+17.6	****0.0000
	PA36	17	1.59	8	0.7	+3.94	*0.0472
	PA37	15	1.4	8	0.7	2.65	0.1034
	Subtotal	54	5.05	19	1.66	+19.89	****0.0000
	Total	4696	439.34	5548	485.82	−25.78	****0.0000

Lastly, it is shown that patterns in Set 5 are more commonly used in pure disciplines than in applied disciplines (LL = 19.89, $p < 0.0001$). The higher frequency of these more complex patterns in pure sciences may be ascribable to the inherently more complex modes of argumentation in these theory-driven disciplines. Statistical significance is evident in PA35 (LL = 17.6, $p < 0.0001$) and PA36 (LL = 3.94, $p < 0.05$). The commonality of these two patterns lies in the cooccurrence of 'Initiator' and 'Source-internal', manifested in the sentence pattern *'by X, we mean Y'*. When enacting these two patterns, the writer first uses a preposition '*by*' to introduce the

Table 12
Example of PA35 from LinDEAP.

Initiator	Definiendum	Source-internal	Hinge	Definiens
By	*'context'*	*I*	*mean*	*'who is asking what and for what reasons. '*

'Definiendum' and then specifies that the definition is proposed by the writers themselves, acknowledging the writers' agency, accountability, and contribution in the process of knowledge construction. An example of PA35 from LinDEAP is given in Table 12.

5. General discussion and pedagogical implications

This study presented a local grammar based investigation into the use of definition across four distinct disciplines. It provided empirical evidence to show that the linguistic and rhetorical features of discourses not only reflect but also shape the values, norms, and epistemological orientations within and across disciplinary communities.

First, this study uncovered notable distinctions between the soft and hard disciplines in their ways of defining. Academic writers in soft disciplines frequently draw on definitions from existing literature, which is reflected by the more frequent use of local grammar patterns of definition characterised by 'Source-external'. Despite the common perception (or stereotype) of soft disciplines as lacking a unified research paradigm and universal standards for validating knowledge due to their interpretative and subjective nature (Becher, 1987, 2012), our findings suggest something different. That is, our study shows that writers in the soft disciplines are aware of the importance of anchoring their personal (re)interpretation or subjective judgement in established definitions or concepts which allow them to build a more reliable knowledge framework, thereby enhancing the credibility and rigor of their work (Charles, 2007; Hyland, 2013). In other words, writers in soft disciplines strive to maintain a balance between subjective engagement and scientism (Glynos & Howarth, 2007).

On the other hand, writers in the hard discipline exhibit a marked preference for the simpler and more prototypical patterns in their definitional practices, particularly those initiated with the element 'Definiens'. Precisely, writers in hard disciplines tend to present 'Definiens' first – the 'Given' information based on the textual context, and then 'Definiendum', which is more abstract, technical, and unfamiliar to the readers, later in the definitional clause as the 'New' information. Structuring definitions in this manner makes the definition clearer, more logical, and more easily understandable, thereby facilitating a gradual progression that assists readers to build a foundation of understanding grounded in their prior knowledge before being introduced to more abstract or complex ideas. This approach likely mirrors the epistemological values of hard disciplines to prioritise simplicity, accessibility, and a cumulative manner of knowledge construction (Becher, 2012).

The comparative analyses across pure and applied sciences also underscored significant disciplinary differences in how researchers communicate their concepts. In pure sciences, there is a heightened tendency for researchers to attribute sources, particularly when presenting self-devised definitions. This finding echoes Hyland's (2001) observation of a considerable increase of self-mentions across pure sciences, suggesting that such authorial intrusion enables the researchers to reinforce their certainty and commitment towards a definition. Moreover, writers in pure sciences tend to apply more complex patterns when enacting definition. This complexity reflects the intricate nature of theoretical constructs involved in these disciplines and the need to articulate nuanced logical relationships, which are crucial to the pure sciences' objective of advancing our understanding of fundamental laws governing social or natural phenomena. Writers in applied disciplines have been found to use the discourse function of definition more frequently than those in pure disciplines but favour simplicity and clarity by resorting to the most prototypical patterns. This preference stems from the practical orientation of applied research, which aims to address real-world issues and enhance societal wellbeing, typically aligned with market and societal needs (Becher & Trowler, 2001). The necessity to communicate effectively with practitioners, policymakers, and the general public requires clear and straightforward explanations that facilitate easy comprehension and direct application of research findings, thereby ensuring relevance and impact beyond the academic sphere.

Overall, these findings highlight the strategic use of definition in academic discourses which is deeply influenced by the objectives, audiences, and values of the disciplinary communities. The findings of this study could have important implications for EAP/ESP writing research and pedagogy. First, methodologically, this study has demonstrated that local grammar, as being able to quantify the realisations of discourse acts at the discourse-semantic level, can offer a novel and robust approach to probing into disciplinary variations of pragma-discursive features of academic discourses, thereby offering an additional toolkit for cross-disciplinary investigations. Second, the findings from this study could usefully inform the local-grammar-based teaching and learning of EAP writing across disciplinary contexts. Prior research has discussed the potential value of incorporating local grammars into EAP writing pedagogy, including the development of teaching materials and course designs with a specific focus on the appropriate performance of discourse acts (Su & Zhang, 2020; Zhang & Su, 2022). We posit here that the instruction on the pragmatic performance of discourse acts would be even more fruitful and effective if the local grammar patterns of discourse acts can be related to discipline-specific knowledge-making practices and epistemological orientations. Specifically, we propose the following measures in implementing a local-grammar-based, discipline-specific EAP writing pedagogy.

1) Compilation of specialised corpora. The instructor can compile, independently or in cooperation with the students, a small corpus of RAs (or any other target academic genres) representing the up-to-date discursive practices of the students' disciplines.
2) Data retrieval and local grammar analysis. The students can be introduced to the significance and core functions of the target discourse act, guided to retrieve instances of the discourse act from the compiled corpora, and then engage in detailed local grammar analyses of the retrieved data.
3) In-depth discussion. Once the local grammar of a given discourse act is established, the students can delve into more profound discussion of the structural characteristics of each pattern, their prototypicality and rhetorical effects in particular co(n)texts, and more importantly, how they contribute to the construction of disciplinary knowledge and relate to the discipline's epistemological values.

4) Comparative analyses. This is an essential part and particularly relevant when dealing with composite classes where students are from diverse disciplinary backgrounds. Students are encouraged to share and compare their local grammar analyses of discourse acts and see how their own discipline-specific writing practices might differ or converge with those of others.
5) Assessment and feedback. Assessment of students' writing samples should not only account for linguistic accuracy but also the appropriateness of discourse acts within disciplinary contexts. Feedback from a disciplinary perspective can also be particularly beneficial.

Such an approach would enhance the capacity of student/novice EAP writers by fostering their critical understanding of disciplinarity and equip them with pragmatic knowledge to skilfully navigate discourse acts in academic writing in accord with the epistemologies of their disciplinary communities. In our future endeavours, we shall further assess the effectiveness of incorporating local grammars of discourse acts into disciplinary writing instruction.

6. Conclusion

This study employed a local grammar approach to investigate variations of the discourse act of definition in research articles across four disciplines, i.e., Linguistics, Education, Physics, and Civil Engineering. The investigation identified 37 relatively frequent local grammar patterns of definition in the four selected disciplines. A subsequent comparison of the distribution of these patterns between the soft and hard disciplines, as well as between the pure and applied disciplines, revealed both similarities and differences in the use of definition across different knowledge domains, which were then attributed to the disciplinary cultures and epistemologies. The findings bring us to a more nuanced understanding of the disciplinary variations in language use at both the lexico-grammatical and discourse-semantic levels, thus offering valuable insights into EAP/ESP scholarship, theoretically, methodologically, and/or pedagogically, as has been discussed in Section 5.

Notwithstanding its contribution to the extant literature on cross-disciplinary research, this study is not devoid of limitations. First, only four disciplines were selected and examined in this study, with a single representative chosen from each of the four disciplinary groupings. Admittedly, this selection could only partially reflect the epistemologies and discursive practices of the respective knowledge domains. Future studies may consider expanding the corpus by diversifying the disciplines included or incorporating balanced, multi-disciplinary discourses within each intellectual cluster. Second, as discussed in section 3.1, we limited our discussion to the levels of disciplines and broader disciplinary groupings, disregarding the intradisciplinary variability. It is thus necessary to further explore if intradisciplinary variation (e.g., computational linguistics *vs* theoretical linguistics) also exists in the use of discourse acts, which may contribute to our understanding of interdisciplinarity and intellectual change and development (Becher, 1994). Lastly, our focus in this study was exclusively on the discourse act of definition. It would be worthwhile for future research to apply the local grammar approach to systematically document the disciplinary variations in the use of other discourse acts (e.g., exemplification, comparison and contrast, cause and effects, etc.). Such endeavours would provide a more thorough and in-depth understanding of the pragma-discursive characteristics of disciplinary discourses, thus enriching the meaning-making repertoire for disciplinary writers.

CRediT authorship contribution statement

Yimin Zhang: Writing – review & editing, Writing – original draft, Project administration, Funding acquisition, Formal analysis, Data curation, Conceptualization. **Huan Wang:** Writing – review & editing, Formal analysis, Data curation.

Statement

The study reported in this paper has not been published previously or submitted for consideration in any other journals. Its publication is approved by all authors, and if accepted, it will not be published elsewhere in the same form, in English or in any other language, without the written consent of the copyright-holder.

Funding

The work was financially supported by Humanities and Social Science Fund of Ministry of Education of China – Grant Number 23XJC740009.

Declarations of interest

None.

Acknowledgements

We thank sincerely the Editor and the anonymous reviewers for their constructive comments on earlier versions of this paper. Any remaining errors are ours.

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